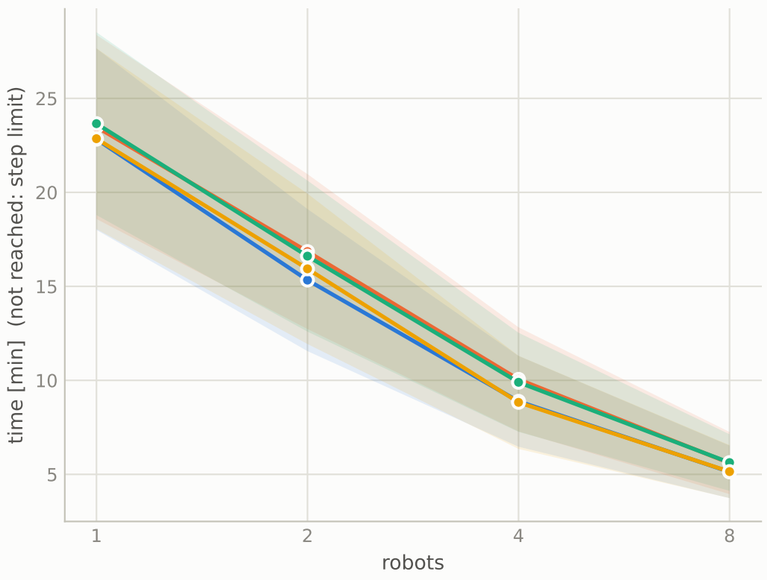
Multi-agent coverage: method comparison

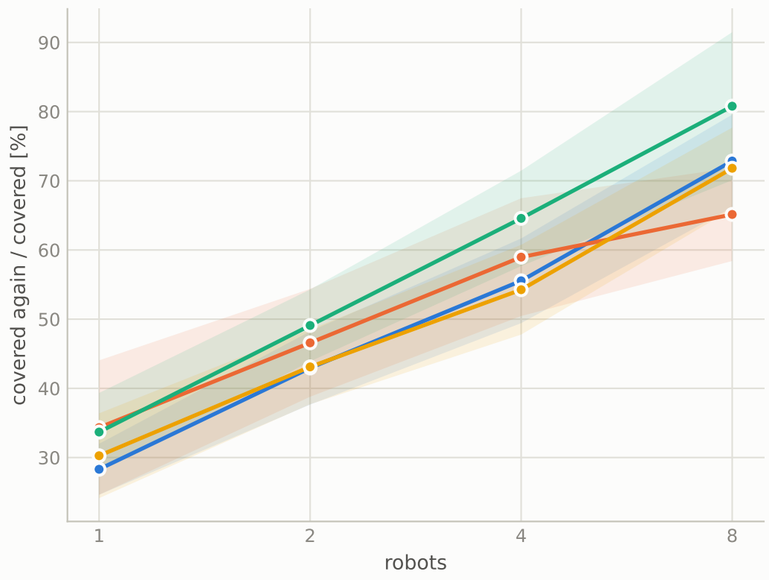
15 maps x 3 episodes per method and team size, deterministic actions, same start poses for every method. Lines: mean, bands: 95 % CI.

MAPPO · IPPO · MAPPO 5 m (comm. 5 m) · MAPPO tested at 5 m (comm. 5 m)

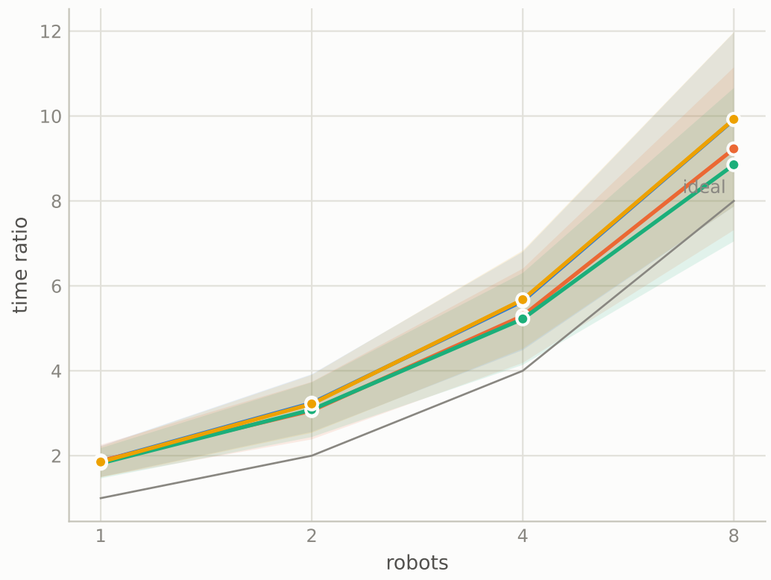
Time to 99 % coverage



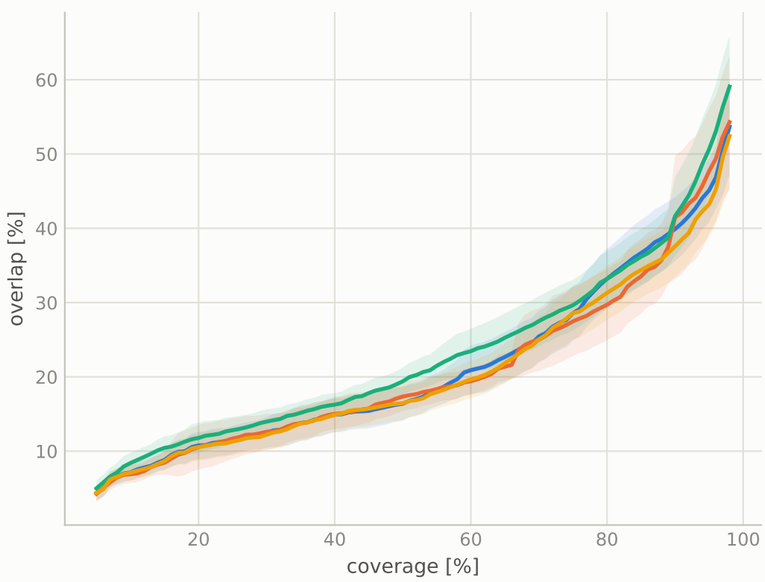
Overlap when 99 % coverage is reached



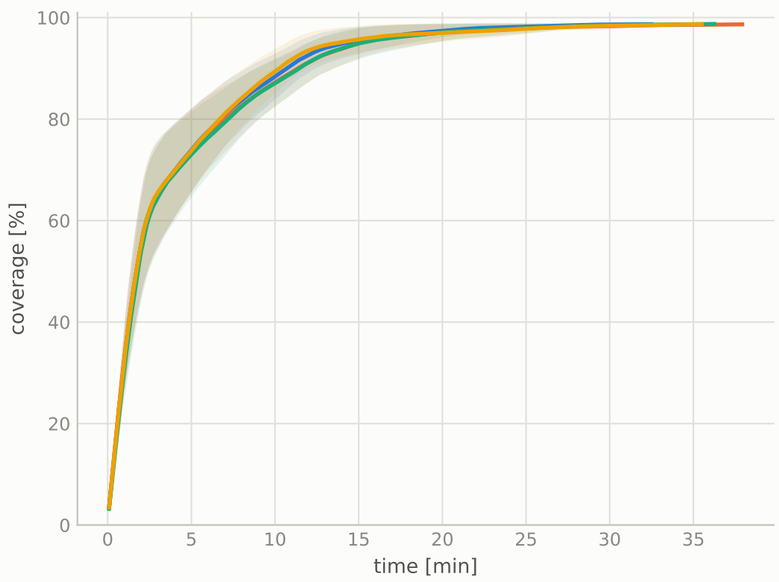
Speedup over 1 robot



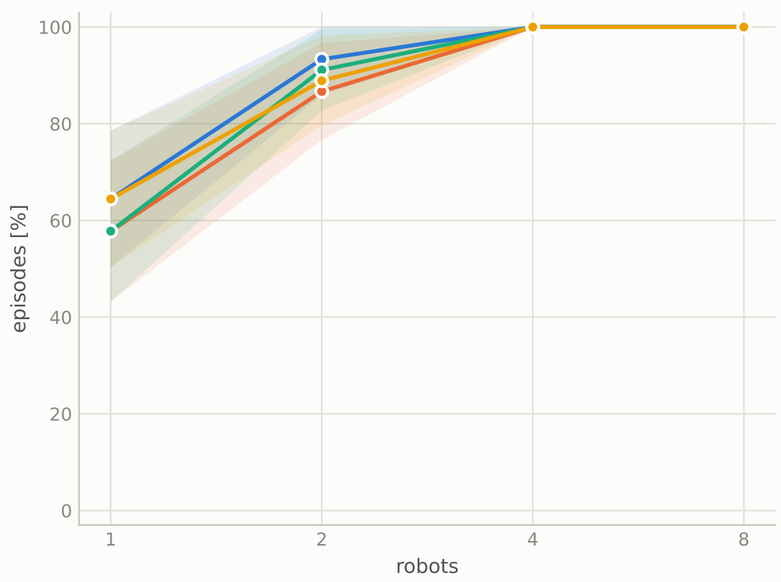
Overlap while covering, 4 robots



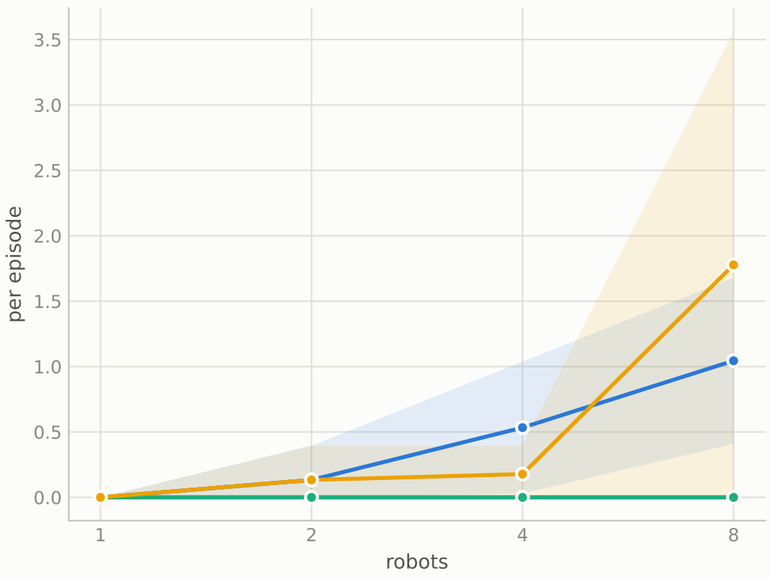
Coverage over time, 4 robots



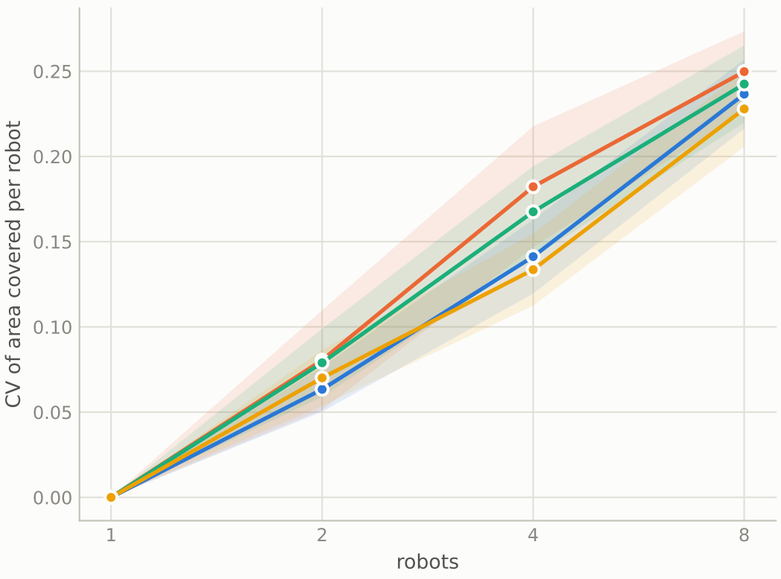
Success rate (99 % coverage)



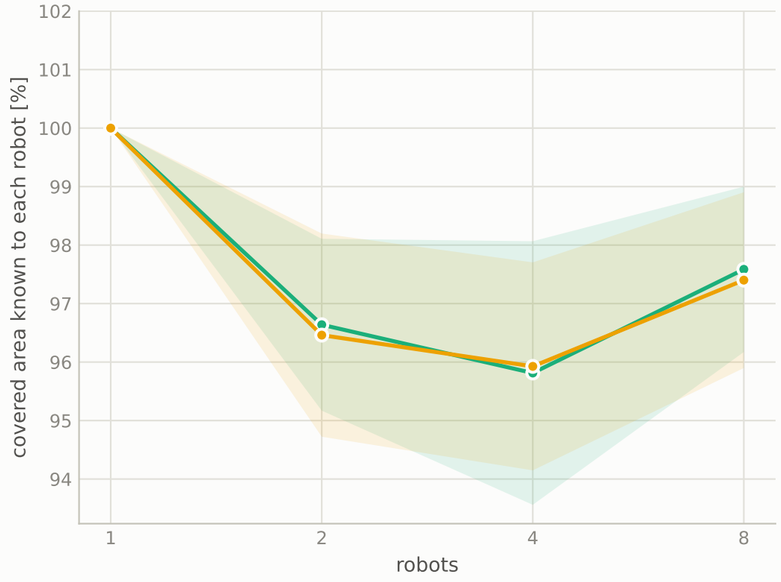
Collisions between robots



Workload imbalance



Belief recall (limited communication)



Summary, 4 robots (mean ± 95 % CI over episodes)

Method	Comm.	Success	Time to 90 %	Time to 99 %	Overlap at 99 %	Final overlap	Robot coll.	Imbalance	Belief recall	Connectivity
MAPPO	unlimited	100 %	7.3 ± 1.9 min	8.9 ± 2.4 min	55.5 ± 6.1 %	55.5 ± 6.1 %	0.5 ± 0.5	0.14 ± 0.02	100 ± 0 %	100 ± 0 %
IPPO	unlimited	100 %	7.9 ± 2.1 min	10.1 ± 2.8 min	59.0 ± 8.5 %	59.0 ± 8.5 %	0.0 ± 0.0	0.18 ± 0.04	100 ± 0 %	100 ± 0 %
MAPPO 5 m	5 m	100 %	7.8 ± 2.1 min	9.9 ± 2.6 min	64.6 ± 6.9 %	64.6 ± 6.9 %	0.0 ± 0.0	0.17 ± 0.03	96 ± 2 %	77 ± 8 %
MAPPO tested at 5 m	5 m	100 %	7.3 ± 2.1 min	8.8 ± 2.5 min	54.3 ± 6.5 %	54.3 ± 6.5 %	0.2 ± 0.2	0.13 ± 0.02	96 ± 2 %	75 ± 9 %